

NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

Project Type

Group / Individual Project

Subject

Cybersecurity Law & Policy

Deliverables

Report + Presentation + Flowcharts +
Infographics

Total Marks

100 Marks

Important Notice to Students

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

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PART 1: Introduction

Prepare an overview section that sets the foundation for your cybersecurity law. This section should clearly articulate why cybersecurity legislation is necessary for VIT's.

What to Include

- Purpose of the law — What is this law designed to achieve?
- Importance of cybersecurity — Why does VIT's need this law now?
- Challenges faced by VIT's — Describe specific threats from the scenario •
- Goals and expected outcomes — What will change after this law is enacted?

Scenario Context

VIT's is a rapidly growing digital nation with 50 million internet users, online banking, e-commerce platforms, government digital services, smart city infrastructure, AI-based public services, and cloud-based businesses. Recently, VIT's experienced large-scale data breaches, ransomware attacks on hospitals, banking fraud, social media misinformation campaigns, identity theft cases, and cyberbullying incidents.

Cyber Law Committee | Academic Project Document | Confidential - For Educational Use Only



PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

Required Terms to Define

#	Term	Definition (To Be Completed by Student)
1	Cybercrime	Cyber crime is a crime which is done digitally using internet , computers, ai or a digital device
2	Hacker	Hacker is a person who keep check on computer networks or systems legally or illegally
3	Ethical Hacker	Ethical hackers means a person hacks or check systems or networks with having official permissions
4	Malware	Malware is a software that damage our system or steal our information or make our system slow
5	Ransomware	Ransomware is a type of malware which locks our files and demands money from us to unlock it
6	Phishing	Phishing is a scam where an attacker or a person pretend to be trusted and try to reveal your sensitive information or otp or passwords
7	Data Breach	Data breach is an incident where sensitive , confidential or personal information is stolen without authorization
8	Personal Data	Personal data is also known as the identification of a person such as DoB name email id etc.
9	Sensitive Data	Sensitive data is a data which need extra protection data like passwords biometric data financial data or health records
10	Digital Signature	Digital signature is an electronic method used to verify authenticity of a digital document
11	Electronic Evidence	Electronic evidence is an information which is transmitted through electronic devices which can be used as proof in legal proceeding
12	Identity Theft	This is an unauthorized use of someone's name and information or doing crime
13	Artificial Intelligence	AI is an intelligence technique which can perform human activities very efficiently
14	Cloud Computing	Cloud computing is delivery of services like storage, software over internet instead of using own device

15	Critical Infrastructure	<i>It refers to the essential system like water transport power communication that are for country safety and also for daily life</i>
16	Cyberbullying	<i>It is the use of digital devices to harass or threaten someone repeatedly</i>
17	Deepfake	<i>It a fake audio video or images created with the help of ai and make it such as it would look like its real</i>
18	Encryption	<i>It is a process of converting data into a secret code so that only selected people can read it</i>
19	Firewall	<i>It controls network traffic and blocks unauthorized access and protects the computer or networks</i>
20	Zero-Day Exploit	<i>It is a cyberattack which uses software that is unknown to the developer and has not fix it yet</i>



PART 3: Citizens' Digital Rights

Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

Required Rights

Right to Privacy

Citizens must be protected from unauthorized collection, use, or sharing of their personal data.

Citizens have right to keep their information private from any unauthorized collection or systems

Right to Secure Data

All organizations must store and protect citizens' data using industry-standard security measures

All the organization must protect citizens personal data using secure systems

Responsibility is of the organization or any agency who collects or store personal data of any person

The data protection authority will conduct an audit and if the organization fails to secure citizens data they can take legal actions against them.

Right to Data Deletion

Citizens may request the complete deletion of their personal data from any organization's systems.

The organization is responsible for securely removing the data and the higher authority can take action if the deletion request is not followed

Right to Information

Citizens have the right to know what data is collected about them and how it is used.

Organizations are responsible for clearly informing users about how their data is used where it is used for what it is used for . the DPA can take legal actions if they are using the information in wrong way

Right to Report Cybercrime

Citizens can report cybercrimes without fear of retaliation, and authorities must act within defined timeframes.

Law enforcement and cyber crime cells are responsible for registering complaints and taking actions within the deadlines. The DPA can take actions if there are delays or negligence of those cases

Right to Fair Investigation

Any person accused of a cybercrime is entitled to a fair, transparent, and timely investigation.

Any person accused of a cybercrime has the right to investigate. Authorities should follow due process and oversight bodies may enter if there is biasness or misconduct or delay occurs

Instruction

You must include at least 6 rights. You may add more. Each right should be at least 2-3 paragraphs with specific implementation details.



PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

Crime	Description + Example	Punishment	Fine
Unauthorized Access	Sneaking into someone's system [Eg. using Erp login id]	6 Month Digital Lockout	50,000
Data Theft	Stealing Student data [downloading student data without permission]	Attend cyber ethics classes with own expenses	20,000
Identity Theft	Using another's id [submitting anothers work pretending its yours]	No submission marks + submitting extra work given by prof.	1000
Phishing	Fake login traps [Fake vit email asking for password]	Mandatory anti phishing training	5,000
Ransomware	Locking systems [locking systems and demand for money]	Lab cleanup duty and no wifi access	5,000
Malware Distribution	Spreading viruses [virus spread via usb in lab]	System recovery session	7,000
Cyberbullying	Online harassment [Mocking someone in class groups online]	Counselling session and behaviour tasks	8,000
Online Harassment	Repeated Abuse [Repeated abusive msg online]	24/7 monitoring period	10,000
Financial Fraud	Fake Payments [Fake exam fee payment links]	Financial ethics seminar	15,500
Misinformation	Fake news spread [false news of exam cancellation]	Fact checking assignment	9,000
Website Defacement	Hacking website [changing college website content]	Website restoration duty	4,000
Cyber Terrorism	Major cyber attack [shut the university network attack]	Legal action + no System access	20,500

PART 5: Data Protection Framework

Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

Data Collection Rules

There are few rules of data collection rules:-

They should collect data for only valid purposes with user consent. They should gather only the necessary information and they should inform users how their data should be used. Sensitive information should be collected and protected with secure passwords. Users should have rights to edit, add or delete the data they want . They should follow the data protection standards.

Data Storage Rules

They are few rules for Data storage Rules:-

They should store data safely so that unauthorized systems cannot look into it. They should store only required data and delete the unnecessary information which they are keeping with them. Regular backup should take place so that data is not misplaced . They should obey the laws for storing the data as per the rules given by higher authority.

Data Sharing Rules

The following are the rules for data sharing :-

They should share only the valid data and which is needed for legal purposes. They should seek permission from users to share the private data. Only necessary information should be shared. They should inform the user which is shared and to whom. The companies should follow the rules which are formed by the higher authority.

User Consent Requirements

Following are the user consent requirements:-

The user should grant permission before the data is collected or used. The consent request should be easy to understand and clear too. The users should have rights to accept or deny the consent. They should have the right to withdraw their consent at any time. The sensitive information should be exposed with consent only. The organization should explain how the user's data is used . they should follow rules while managing the user consent.

Data Retention Period

Following are the Data retention period :-

The user's data should be kept until it's needed for the purpose. Once the purpose is done the data should be deleted or demolished. They should mention a retention timeline for data . The sensitive data should have small retention time. Regular checks should take place to ensure the

unnecessary data is deleted .

Data Deletion Process

Following are the data deletion process rule :-

Users can request the organization to delete their data at any time. Only the requested data should be deleted . Users should be informed once their data is deleted . The data should be removed from all the active data bases and systems companies should maintain the logs for their organization reports . The organization should follow legal and data protection guidelines.

Breach Notification Requirements

Organizations must quickly identify the data breaches. Affected users should be informed without any delay. The message should clearly mention which data has been exposed or passed to them. The user should be aware of possible risks of it. They should also provide steps to protect their information and data. Relevant authorities must be informed by law and orders . companies should take preventive measure to prevent further damage.

PART 6: Corporate Responsibilities

All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

Required Responsibilities (Minimum 10)

Responsibility 1: Use Secure tools for security

Must use firewalls, antivirus and secure software

Fine up to 78,000

Responsibility 2: Regular security check

VIT should conduct cybersecurity audits once or twice a year

Fine up to 50,000 and mandatory check should take place

Responsibility 3: Train the computer cell department

The dept should get cybersecurity training twice or thrice a year

Fine up to 60,000

Responsibility 4: Must report cybercrime earliest if it occurs

Must report the authorities as soon as possible

Fine for delaying to report the crime

Responsibility 5: Protect Student n faculties data

Data of student and teaching staff should be protected and keep confidential

Fine compensation and legal punishment

Responsibility 6: Strong password policy

All the faculty and student should keep strong pass and should not share there credentials with others

Mandatory password security workshop and warning for the first time

Responsibility 7: Regular updates of each software

Devices used on campus network should be updated regularly

No network access until the updates take place

Responsibility 8: Secure data backup

Department should have backup for the important academic data

Fine 5,000 and Extra documentation work

Responsibility 9: Restricted use of cloud services

Staff should use approved cloud platforms for storing files

Removal of unauthorized files and mandatory cybersecurity training

Responsibility 10: Data Sharing

Every VIT authorities should not share confidential and secure file to the other unauthorized people

Disciplinary actions and cyber ethics assignment

Examples to Consider

Maintain security controls | Conduct regular security audits | Train employees on cybersecurity | Report incidents within 72 hours | Protect customer information | Appoint a Data Protection Officer | Conduct annual penetration testing | Implement access control policies | Encrypt sensitive data | Maintain incident response plan



PART 7: Government Responsibilities

Specify how government agencies of VIT's must protect national interests in cyberspace. Include concrete mechanisms for each area.

Protect National Infrastructure

VITs infrastructure consists of an ERP system , database , examination portal, campus wifi, etc.

All the systems of VIT should be secure with the use of firewalls ,software update, regular check and strong passwords . if any inconvenience occurs immediately report to IT department for quick action and recovery.

Monitor Cyber Threats

The IT Department of VIT's should regularly monitor the college website, ERP system, Wi-Fi network, and student database for any suspicious activities. Security software and alerts should be used to detect threats early. Any unusual activity must be investigated immediately to keep college data and systems safe.

Investigate Cybercrimes

If a cybercrime occurs, the IT Department should collect digital evidence and investigate the incident fairly. Serious cases should be reported to higher authorities. Proper records must be maintained to identify the person responsible and prevent future incidents.

International Coordination

VIT's can work with cybersecurity experts, educational institutions, and law enforcement agencies to share knowledge and learn about new cyber threats. This helps improve the college's cybersecurity practices and response capabilities.

Awareness Programs

VIT's should conduct regular workshops, seminars, and awareness campaigns to teach students and staff about cyber safety, phishing, password security, and responsible internet use. This will help create a safer digital environment on campus.



PART 8: Cyber Incident Reporting Framework

Design a step-by-step process for how cyber incidents are reported, investigated, and resolved in VIT's. You must also create a flowchart (as a separate deliverable).

Reporting Process Steps

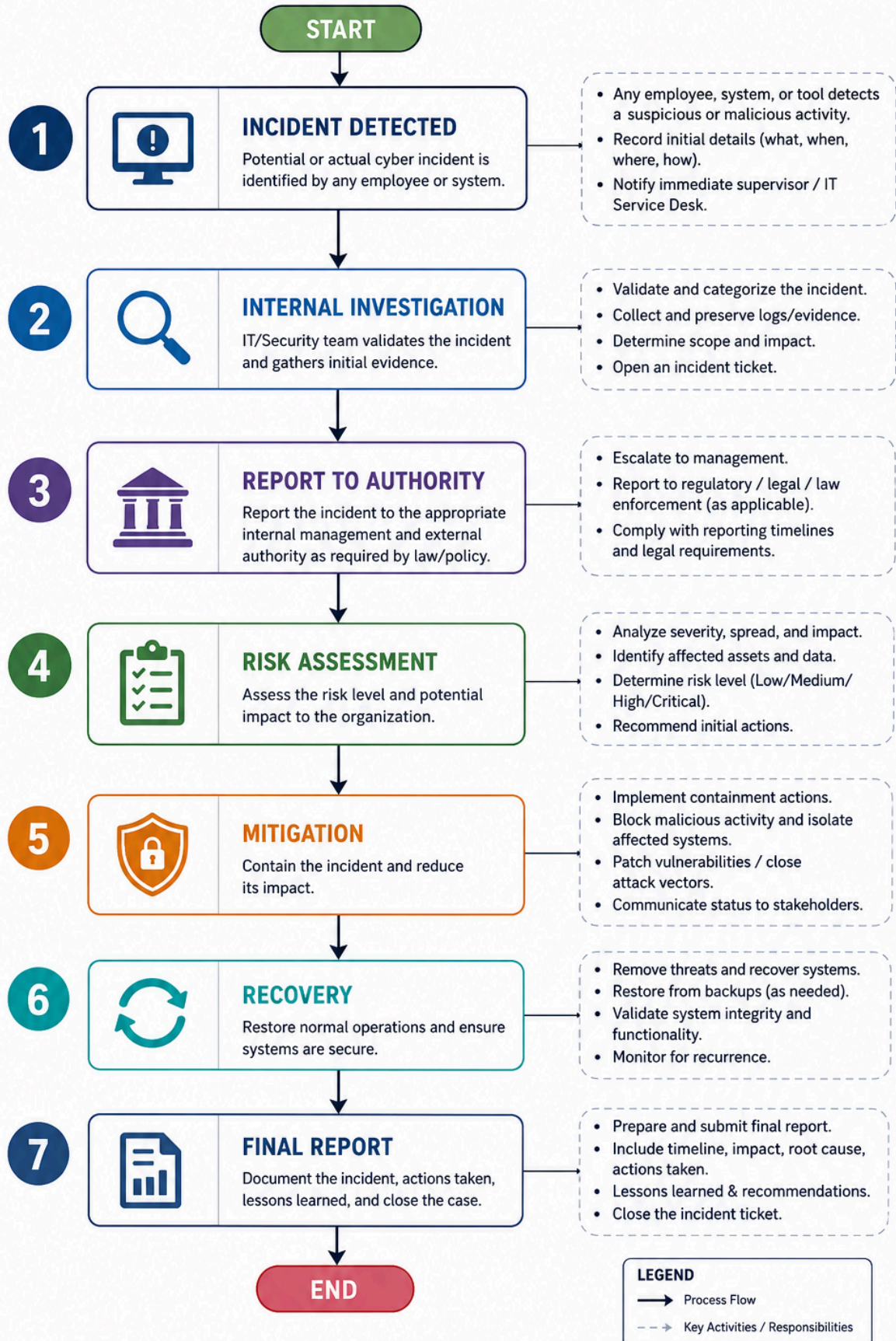
Step	Stage	Description (Student to Complete)
1	Incident Detected	The moment any employee, system alert, or automated monitoring tool notices something unusual like a sudden system slowdown, unauthorized login attempt, or suspicious file activity it must be immediately reported to the organization's internal IT security team. This must happen within 1 hour of detection and no employee is allowed to ignore or delay reporting even if they are unsure whether it is a real attack or not. Quick detection is the first and most important step in reducing damage.
2	Internal Investigation	Once the IT security team is informed they must immediately begin an internal investigation to understand what happened, which systems were affected, and how serious the situation is. This stage must be completed within 6 hours and the team must document everything they find including which files were accessed, which accounts were used, and what time the suspicious activity began. No affected system should be switched off or wiped during this stage as it could destroy important digital evidence.
3	Report to Authority	After the internal investigation the organization must formally report the incident to the NCSA within 6 hours for critical sectors like banking, healthcare, and energy and within 24 hours for all other organizations. The report must include details of what happened, which systems were affected, how many citizens or records were impacted, and what steps have already been taken. Delaying or hiding a cyber incident from the NCSA is itself a punishable offense under VIT's cybersecurity law.
4	Risk Assessment	Once the NCSA receives the report its Incident Response Team conducts a thorough risk assessment to determine how serious the attack is and what further damage could occur if action is not taken immediately. This assessment looks at whether the attacker is still inside the system, whether sensitive citizen data has been stolen or exposed, and whether the attack could spread to other connected systems. Based on this assessment the NCSA decides the level of emergency response required and assigned the appropriate resources
5	Mitigation	At this stage the main goal is to stop the attack from causing any more damage. The NCSA's Incident Response Team works alongside the organization's IT team to isolate compromised systems, block the attacker's access points, remove malware, and patch the vulnerabilities that were exploited. If the attack involves a website or platform it may be temporarily taken offline to prevent further harm to citizens. All mitigation actions must be

		documented carefully so they can be reviewed later during the final report stage.
6	Recovery	Once the threat has been fully contained the focus shifts to restoring all affected systems and services back to normal operation as quickly as possible. Data is restored from clean backups, all security patches are applied, and every system is thoroughly tested before being brought back online to ensure the attacker cannot re-enter through the same weakness. Citizens and organizations affected by the incident must be informed about what data was compromised and what steps they should take to protect themselves. Recovery must be completed within a reasonable timeframe depending on the severity of the attack.
7	Final Report	After everything is back to normal the organization must prepare a detailed final incident report and submit it to the NCSA within 30 days. This report must cover what exactly happened, how the attack occurred, what damage was caused, how it was resolved, and most importantly what changes have been made to prevent the same thing from happening again. The NCSA reviews all final reports and uses the findings to update national cybersecurity guidelines so that every organization in VIT's can learn from each incident and become better prepared for future attacks.

Deliverable Reminder

You must submit a separate FLOWCHART showing this complete process. Use tools like Canva, draw.io, Lucidchart, or PowerPoint SmartArt to create a professional diagram.

CYBER INCIDENT REPORTING FRAMEWORK





PART 9: National Cyber Security Authority (NCSA)

Design the organizational structure, responsibilities, and powers of the NCSA — the primary government agency responsible for cybersecurity in VIT's.

Organizational Structure

The NCSA is the top government body responsible for all cybersecurity matters in VIT's and it operates directly under the Ministry of Digital Affairs. At the top is the Director General who oversees the entire organization and reports directly to the Minister. Below the Director General there are four Deputy Directors each responsible for a specific area namely operations, policy, legal affairs, and public education. Each Deputy Director manages their own team of specialists and all departments report upward through a clear chain of command ensuring quick decision making during cyber emergencies.

Core Responsibilities

The NCSA is responsible for protecting VIT's entire digital landscape from individual citizens to critical national infrastructure. Its main functions include creating and updating national cybersecurity policies, monitoring cyber threats across government and private networks, investigating cybercrime complaints, certifying cybersecurity professionals and organizations, conducting public awareness campaigns, and coordinating with international cybersecurity agencies. The NCSA also reviews and approves cybersecurity standards that all companies operating in VIT's must follow and publishes an annual national cybersecurity report for public transparency.

Legal Powers

The NCSA has strong legal powers to carry out its responsibilities effectively. It can compel any company or government department to share data, system logs, or security records during an active investigation. It has the authority to shut down websites, apps, or online platforms that pose a threat to national security or citizen safety. It can issue emergency orders to freeze suspicious financial transactions linked to cybercrime. It can conduct raids on premises suspected of being used for cybercriminal activities with proper court authorization and it can impose fines on organizations that fail to meet cybersecurity compliance standards without needing lengthy court proceedings.

Investigation Process

When a cybercrime complaint is received the NCSA follows a clear step by step process. First the complaint is registered and assigned to the Cybercrime Investigation Unit within 24 hours. Second investigators collect all available digital evidence including logs, emails, IP addresses, and device data. Third, the evidence is sent to the Digital Forensics Lab for detailed technical analysis. Fourth if the case involves a foreign country the International Cooperation Unit is activated to request assistance from partner agencies. Fifth suspects are identified and arrest warrants are obtained through proper legal channels. Finally all evidence and findings are compiled into a detailed report submitted to the court for prosecution ensuring the case is handled transparently and legally from start to finish.

Departments

The NCSA has five specialized departments each handling a specific area of cybersecurity. The first is the Threat Intelligence Unit which continuously monitors global and domestic cyber threats and issues early warnings to government agencies and critical sector organizations. The second is the Incident Response Team which is deployed immediately whenever a major cyberattack occurs and is responsible for containing the damage and restoring affected systems as quickly as possible. The third is the Digital Forensics Lab which handles all technical investigation work including analyzing malware, recovering deleted data, and preparing digital evidence for court proceedings. The fourth is the Policy and Compliance Department which creates cybersecurity standards, conducts audits of organizations, and ensures that all companies in VIT's are following the law. The fifth is the Public Awareness and Education Department which runs nationwide cybersecurity campaigns, organizes workshops, and works with schools and colleges to build a cyber aware society across VIT's.



PART 10: AI and Emerging Technology Regulations

Create specific cybersecurity regulations for emerging technologies used in VIT's. For each technology, identify key risks and write enforceable rules.

Artificial Intelligence

Key Risks

Artificial intelligence is becoming common in areas like healthcare, banking and government services. If attackers manipulate AI system with False information the AI may make incorrect decisions. AI can also be misused to create fake content, automate cyberattacks and spread misinformation on a large scale.

Regulations

AI system used in important public services must be tested and approved before deployment. Human supervision is required for major decisions such as medical diagnosis or loan approvals. using ai for cyber crime, phishing, Malware Creation or harmful defect will be treated as a serious criminal offence developers must regularly report security issues and data source to the NCSA.

Deepfakes

Key Risks

Deepfakes can create realistic fake videos and audio recordings that are difficult to identify; they can be used to spread false news, impersonate officials, damage reputations and influence public opinions. Such content can spread quickly and create confusion among citizens.

Regulations

Anyone creating or sharing harmful deep fakes With the intention to deceive other can face legal action . social media platform must remove reported deepfake content quickly. AI-generated media should be clearly labelled And the NCSA will monitor and take action against harmful deep fake activities.

IoT Devices

Key Risks

Smart devices such as security cameras, Smart TV And fitness trackers are often connected to the Internet. If this device are not properly secured, Hackers can access personal information, spy on Users or use the device in large cyber attacks.

Regulations

All lot devices must pass cybersecurity checks before being sold. Manufacturers must provide strong passwords and regular security updates. User should be informed about what data is collected and how it is used. Devices involved in cyber attacks may be banned from the market.

Cloud Computing

Key Risks

Cloud services helps organization store add manage data online, but they also create security concerns. Sensitive information may be exposed through hacking, services outages, Or weak security controls. A signal cloud breach can be affect many organization at once.

Regulations

Organization handling sensitive data must be approved cloud services that meets security requirements. Cloud provider must follow standard such as encryption and regular audits. Data breach incident must be reported immediately and organizations should maintain backup and recovery plans.

Digital Identity Systems

Key Risks

Digital identity systems make it easier for citizens to access services online, But they also become attractive targets for cyber criminals. If digital identities are stolen criminal may commit fraud, access personal accounts or misuse govt services.

Regulations

Digital identity systems must be use strong security Measures such as encryption and multi

factor authentication. Citizens should have control over their personal information and be able to track its usage. Unauthorized access or misuse of digital identities will result in strict penalties.

Autonomous Systems

Key Risks

Autonomous technologies such as self driving cars, drones and industrial robots depend heavily on software and network connectivity. If hacker gain control of the systems they could cause'cause accident, property damage or injuries.

Regulations

All autonomous system must pass cybersecurity certification before use. A manual override option should always be available in emergencies. Manufacturer must provide regular security updates and quick fix vulnerability. Cyber attacks cause physical harm will attract severe punishment under cyber security laws.

Infrastructure Protection

Create sector-specific cybersecurity requirements for each of the following critical infrastructure types in VIT's. Requirements should be practical and technically grounded.

Sector	Cybersecurity Requirements (Student to Complete)
Banks & Financial	Banks in Vit's Handle crores of rupees every single day making them the biggest target for cyber criminals. All banks must used multi factor authentication so that password alone is never enough to access any account. Customer financial data must be stored in encrypted form To prevent Misuse even if the hackers break into the system. Banks must conduct penetration testing twice a year to fine and fix witnesses before real attackers can exploit them. A dedicated cybersecurity team must monitor all transactions In 24 hours a day And reporting anything suspicious immediately. Any data breach must be reported to the NCSA within 6 hours an affected customer must be informed within 24 hours. Banks must also train all their employees on cyber security at least twice a year so the human error is a very common cause of attack. all banking software must be regularly updated and outdated system must be replaced without any delay. Banks must appoint a chief information security officer who is directly responsible for all the overall digital security of these institutions. Any bank that fails to follow this rule will face heavy fines and suspension of its digital banking licence until oil issues are fixed.
Hospitals & Healthcare	As hospital store highly sensitive patient data and a cyberattack here can directly risk human lives as seen in the Vits Recent ransomware attack on hospital systems. To prevent This all hospitals must store patient data in encrypted database separate from general office networks, Maintain daily offline backup of critical data Secure all Internet connected medical device with regular updates and strong passwords, appoint a dedicated IT Security officer to monitor system continuously and report any cyber attack to the NCSA Within four hours without fail. Hospitals that do Compliance will face strict penalties and mandatory government security supervision.
Airports & Aviation	Airports depend on hundreds of digital systems simultaneously and Any cyber attack can create serious safety risks for thousands of passenger travelling through VIT's. To ensure safety all airports must operate air traffic control system on completely private networks with no public Internet connection provides cyber security training to all staff every six months, Maintain real time backups of all critical systems, Conduct annual cyber security stimulation drills and report any suspicious activity to the NCSA within 2 hours of detection. Airports failing to meet these standards will face heavy fines and mandatory shutdown of compromised system
Railways & Transport	Railway in VIT's carrying millions of passenger daily and a cyber attack on signalling system directly lead to train accident and loss of life. To

	Prevent this all railways must operate Signalling and control system on completely isolated networks with zero Internet Connection, Use encrypted connection for all digital ticketing platform, Conducts cyber security audit every 6 months And submit Result to NCSA , Train all control room staff to detect and respond to cyber attacks and ensure all software is regularly updated with the latest security patches. Railway that failed to comply Will face heavy fines and direct NCSA intervention.
Power Plants & Energy	Power plants are the backbone of VIT's and if taken down by a cyber attack the entire country including hospitals and communication networks will stop functioning. To protect this critical sector all power plants must be kept SCADA an industrial control systems Completely air gapped with no Internet connection, conduct thorough background checks and annual cyber security training for all staff with system access, maintain fully functional backup system for immediate takeover during any attack, Share threat intelligence regularly with the NCSA to warn Other plans of emerging threat And source hardware and software only from government approved vendors to prevent Supply chain attacks. Non compliant Energy companies will face the highest possible fine and immediate government takeover of security operations.
Government Data Centers	Government data centre holds the most sensitive information in VIT's ranging from citizen identity records to national security files and any breach would be a disaster for the entire country. To prevent this facilities all government data centres must be strictly control physical access using biometric authentication and round the clock security guard, encrypt all stored data with key maintenance separately from the data itself instal real time monitoring system to immediately detect and alert any suspicious network activities, Maintain complete data Backup at a separate geographic location for quick recovery and undergo an independent 3rd party security audit every year with result submitted to both the NCSA and a parliamentary oversight committee. Govt department Found negligent Will face serious administrative action and mandatory remediation under full NCSA supervision.



PART 12: Awareness and Education

Design a National Cybersecurity Awareness Campaign for VIT's. Your campaign must reach all segments of society through age-appropriate and effective programs.

School Programs (Age 6-18)

Teaching cyber security from a young age is one of the smartest thing vits can do for its future.

Children today are using smartphones and the Internet from a very early age so it makes sense to teach them how to stay safe online right from school. For younger kids between age 6 to 10 the focus should be on the basic things like not sharing personal information with strangers online, understanding what a safe website looks like, and telling a trusted adult if something feels wrong on the Internet. These lessons can be taught through fun activities like games, cartoons and storytelling so that children enjoy learning without feeling bored. For middle school students between age 11 to 14 the topics should include things like password safety, recognising fake news, understanding cyber bullying and what to do if they face it. For high school students between age 15 to 18 the curriculum should cover more serious topics like data privacy, phishing attacks, safe social media use, and add basic ethical hacking concepts. Schools should conduct at least one cyber security awareness event every year like a quiz competition or poster making activity to keep students engaged and interested in the subject.

College & University Programs

At the college level students should get a much deeper understanding of cyber security because many of them will go on to work in technology, law, business or government. Every college in Vits should include at least one compulsory cybersecurity subject regardless of what the student is studying. For engineering and computer science students there should be dedicated cyber security lab sessions where they get hands on practice with real tools used in the industry. Universities should also offer cybersecurity certification courses that students can complete alongside their regular degree. This certification will help students to get better jobs and also ensure that Vits has a strong pool of trained cyber security professionals in the coming years. Colleges should also organise annual ethical hacking competitions called capture the flag events where students can test their skills in a safe and legal environment. Guest lectures from industry experts and NCSA officials should be arranged regularly so students understand how cyber security works in the real world and not just in the textbook.

Corporate Training Programs

As it is not enough to only train students because the biggest security risk often comes from employees inside organisations who unknowingly click on phishing emails or use weak passwords. Every company operating in Vits must make cybersecurity training compulsory for all employees regardless of their job role. This training should be done at least twice a year and not just once during joining. The training should cover practical topics like how to identify a phishing email, how to create a strong password, what to do if you suspect a cyber attack and how to handle sensitive customer data responsibly. After every training session employees should take a short test to check if they actually understood what was taught. Companies should keep record of who completed the training and submit record to the NCSA during annual compliance checks. Senior management and IT staff should go through more advanced training that covers topics like incident response, data breach handling, and security audit procedures. Any company that skips this training or provides fake records should face heavy fines.

Social Media Campaigns

A large portion of VIT's 50 million Internet users Spend a significant amount of their time on social media platforms. This makes social media one of the most powerful tool to spread cyber security awareness across the country Quickly and at the low cost. The government through The NCSA run regularly awareness campaign On platforms like Instagram, Youtube, Facebook And and Local messaging apps. The content should be Short, Simple And visual attractive So that people actually stop And read or watch it Instead of scrolling. Short videos of 60 to 90 seconds Explaining Common threats like phishing, fake job offers And online scam Would work well especially for younger audience. Infographic showing the Five mistakes people make online Or how to spot a fake website can be shared widely and easily understood even by people with limited education. The campaign should run throughout the year with special focus during national cyberity awareness month. As influencer and popular content creator should be partnered With To reach audience that may not follow government pages directly. all content should be available in multiple local language So that no citizen Is left Cut due Language barrier.

Public Awareness Workshops

Not every citizen in VI T's Is comfortable using the internet For understanding the risk involved. older citizens, people in Rural areas And small business owners Are especially vulnerable To online scam and fraud Because they are often Have very little knowledge about cybersecurity. To reach this group the government should organized Three public awareness workshop in communities hall ,Libraries ,School And local government offices Across the country. This workshop should be held at least once every three months in every district of VIT's. The topic covered should be practical and relevant to the daily life such as how to avoid online banking fraud, How to recognise a fake phone call Or messaging asking for OTP, How to shop Safely And how to protect personal photos and information on a phone. The session should be kept simple with demonstration and real life examples rather than technical jargons. Local police Officers, NCSA volunteers and trained college students Can be used to conduct this workshop so that more sessions can happen simultaneously across the different location. Participants should be given simple printed guidelines in local languages that they can take home and refer to later. Feedback form should be collected after every workshop So the content can be improved based on what people found useful or confusing.



PART 13: International Cooperation

Define how VIT's will collaborate with other countries and international organizations to combat cybercrime that crosses national borders.

Cybercrime Investigations

Cybercrime does not stop at Borders and many attacks on VIT'S system come from criminal sitting in other countries making it impossible to handle them alone. VIT'S will follow the principles of Budapest Convention on Cybercrime which is most widely accepted international framework for cross border cybercrime operations. When a cyber crime case involves a foreign country VIT'S NCSA will formally contact that country's cybersecurity agency and request assistance in tracking the attackers, collecting digital evidence and sharing investigation findings. All such request must be processed within 30 days and VIT'S will offer the same cooperation to other countries when they need help with case linked on VITS. This mutual cooperation ensures that criminals cannot simply hide in another country and escape justice.

Info Vits will followon Sharing Agreements

One of the most effective ways to fight cyber crime internationally is to share information about new threats, attack methods and criminal groups before they cause damage. VIT'S will sign formal information sharing agreement with friendly nations and international organizations like INTERPOL and the UN cyber Crime Divisions. This agreement will allow VIT'S NCSA Two regularly exchange details about newly discovered malware, hacking techniques and unknown

cyber criminal networks with partner countries. All shared information must be handled confidentially and use only for cyber security purposes and not for political or commercial gain. VIT'S will also join an international cybersecurity information sharing platform so that real time threats data can be exchanged weekly during active cyber attacks.

Joint Operations

Some cyber crimes networks are so large and sophisticated that no single country can be taken them down alone and joint international operations become absolutely necessary in such cases. VIT'S will actively participate in a joint cyber Crime operation organized by INTERPOL, Europol and the author international agencies were multiple countries work together to simultaneously arrest members of criminal networks across different locations. VIT'S NCSA will assign a dedicated international operations unit that works directly with foreign agencies during such joint mission. Before any joint operation begins a formal agreement must be signed between all participating countries clearly define each country's role the legal boundaries of the operation and how evidence and arrest will be handled. Varieties will ensure that all joint operations are conducted in the full compliance with both VIT'S domestic law and international human rights standards.

Extradition Requests

When a cyber criminal who has attacked VIT'S system is found hiding in another country's VIT'S must have a clear legal process to bring that person back to face the justice. VIT'S will sign bilateral extradition treaties with as many countries as possible specifically covering a cyber criminal offense so that attacker cannot use national board as a shield. Then the VIT'S receive a confirmed identity of a cyber criminal in another country the NCSA will work with the Ministry of Justice to file a formal extradition request supported by solid digital evidence collected during the investigation. Similarly when another country requests extradition of a person in vit suspected of cyber crime extradition VIT'S will review the request carefully and respond within 60 days All extradition cases involving cyber crime must be handled with urgency since digital evidence can disappear quickly and delays can allow criminals to destroy important proof.

Threat Intelligence Exchange

Knowing about a cyber attack before it happen is far better than trying to response after the damage is done and this is exactly what threat intelligence exchange is designed to achieved. VIT'S will establish a national threat intelligence sharing network that connects the NCSA with cybersecurity agencies of partner countries allowing real time exchange of information about emerging threats, new attack patterns and vulnerable systems. VIT'S will follow the Traffic Light protocol which is an internationally recognized system that labels shared threat information by sensitivity level so that each country knows exactly how they are allowed to use and share the information they receive. When VIT'S detect a new type of cyber attack or malware it will immediately share technical details with partner country so they can protect their system before being hit. In return partner countries will do the same for VIT creating a global early warning system that makes everyone safer and more prepared against rapidly evolving cyber threats.



Studies

Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

Case Study 1: [The Silent Breach– VIT's National Health Database hack]

Crime Category	Unauthorized Access / Data Breach
What Happened	In March 2024 hackers broke into VIT'S National Health database storing records of 30 million citizens using SQL injection technique. They stayed hidden for 3 weeks copying and selling patient data on the dark web. The breach was only discovered when a journalist reported private medical records are being sold online.
Law Violated	Attacker violated Section 4 covering unauthorized access, Section 7 covering data theft and illegal sales of personal data and section 5 covering failure to maintain adequate security and reports the breach on the time.
Investigation Process	NCSA's Digital Forensic Lab Traced the attack through server logs and identified IP addresses used during the breach. The attack was traced to two foreign countries and an extradition request was filed. 3 suspects were identified within 45 days and 1 was arrested inside VIT'S
Punishment Given	The main attacker receives an 8 year imprisonment and a fine of ₹25,00,000. The health department officially faces suspension and a departmental fine of ₹2,00,00,000. The organization was ordered to overhaul its cyber security infrastructure within 6 months.
Lessons Learned	This case made quarterly security audits compulsory for all

	healthcare institutions. Mandatory encryption for health databases was introduced and breach reporting time was reduced from 72 hours to 4 hours. A dedicated healthcare cyber security division was created within the NCSA.
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Case Study 2: [Operation Fake Pay– Mass banking phishing fraud]

Crime Category	Financial Fraud / Phishing
What Happened	In July 2024 citizens received fake messages pretending to be from VIT National Bank asking them to verify account details through a fake website. Within 72 hours attackers stolen ₹18,00,00,000 from over 12,000 citizen accounts. The scam was discovered when hundreds of customers reported unauthorized transactions to the real bank.
Law Violated	Criminals violated Section 6 covering phishing and online deception, section 9 covering financial fraud and unauthorized banking access and Section 12 covering identity theft for using stolen credentials to access victim accounts illegally.
Investigation Process	NCSA traced all fraudulent transactions within 24 hours and froze 40% of stolen funds before they could leave VIT'S banking system. The fake website was taken down within six hours. Two key suspects were identified within 30 days and arrested through a joint international operation with a neighboring country.
Punishment Given	Main criminals each receive six year imprisonment and fine of ₹50,00,000 each. They were ordered to repay the full ₹18,00,00,000 stolen. Other gang members receive three years of imprisonment and VIT'S National Bank receives a warning notice to upgrade its fraud detection system within 90 days.
Lessons Learned	This case made it mandatory for all banks to implement email and SMS authentication protocol so citizen can verify genuine bank communications. The NCSA launched a nationwide campaign called Think Before You Click to educate citizens about identify phishing attempts and staying safe online

Case Study 3: [Operation Dark Grid– Ransomware attack on VIT's power grid]

Crime Category	Ransomware / Cyber Terrorism
What Happened	In November 2024 state sponsored hackers launch a ransomware attack on Vit's National Power Grid after filtering the system through a phishing email sent to a power ministry engineer three months earlier. The attack caused an 11 hour nationwide blackout affecting hospitals, traffic system and communication networks. Attackers demand 150,00,00,000 in cryptocurrency within 48 hours threatening to

	permanently destroy all control systems.
Law Violated	Attackers violated section 5 covering ransomware attack, section 15 covering attack on critical infrastructure, section 16 covering cyber terrorism and section 4 covering unauthorized access to government system. The power Ministry engineer was also found guilty of Negligence under section 6 of Corporate responsibility
Investigation Process	Govt declared a national cyber emergency and NCSA deployed its incident response team to all 14 affected control centers within two hours. Within 72 hours the malware was traced to a foreign state sponsored hacking group. VIT's activated International corporations agreement and share evidence with INTERPOL. No ransom was paid and power was restored within 11 hours using a backup system.

Punishment Given	Vit's invoked its cyber security warfare policy and raise the matter at the United Nations while imposing diplomatic sanctions on the responsible nation. The negligent engineer received two years imprisonment and a fine of ₹10,00,000. The Power Ministry was fined ₹10,00,00,000 and order to rebuild its cyber security infrastructure within three months under NCSA supervision.
Lessons Learned	The attacker made an air gapping of all power grid control system compulsory across vit's. Monthly cyberSecurity drills become mandatory for all power sector employees. A dedicated critical infrastructure protection force was created under the NCSA and Vit's Accelerated plans to build a National Cyber Defense Command to handle future state sponsored attacks.



PART 15: Future Recommendations

Based on your research, suggest forward-looking improvements and additions to VIT's cybersecurity law for the next 10 years. Consider how technology will evolve and what new threats will emerge.

AI Security (2025-2035)

Over the next 10 years, artificial intelligence is going to touch almost every part How Vit's functions- From banks to government offices and city infrastructure.. That kind of growth brings real opportunities, but it also brings serious risks that we can't afford to ignore. Vit needs to set

up an AI safety board, understand the NCSA that reviews high risk AI systems before they are allowed into sensitive sectors. Any AI used in government decision - making Should always be required to explain its decision in plain language by 2027 - Because citizens deserve to understand why an algorithm affects their life.

At the same time, hackers are already using AI to launch smarter, faster attacks-Realistic phishing emails , self Adapting Malware and manipulating tactics that are harder than ever to catch. By 2028,The NCSA needs its own AI-powered defence platform that can detect and respond to this thread automatically, because waiting for a human to notice is no longer fast enough. There's also the quieter issue of AI bias- Algorithms used in policing , banking can quietly discriminate against certain groups without anyone intending it. VIT'S must make bias audits of all public sector AI system legally mandatory by 2026, not just a guideline that organization can choose to ignore.

Quantum Computing

Right now, Vits use standard encryption to protect bank transfers, government files, and citizen data. But quantum computers, which are being developed globally, will soon be powerful enough to break this encryption methods in minutes. This is a massive risk because all the data that is encrypted today could be decrypted tomorrow once quantum technology matures.

VIT's Should start preparing now by planning a shift to quantum- safe encryption. By 2027, every government department should list all the system they used for encryption and submit a plan to upgrade them. By 2030, all the critical system like banking and defense must used new quantum resistance algorithms. Private companies should complete this shift by 2033. Any organisation that delays without a valid reason should be fined heavily. VIT's should also invest in buildings its owns quantum cyber security research centre to stay .

Privacy Protection

As VIT's grows in a smart nation, more and more data is being collected about citizen everyday -from fitness trackers and Smart tvs to shopping apps and City cameras. The current law covers basic personal data , but it also does not fully address biometric data like fingerprints or face cans or behaviour data like how you can move around the cities or what you buy online. These types of data are very sensitive and need stronger protection.

As VIT should pass a stronger privacy law by 2028. Biometric data should only be collected if the person gives clear and voluntary permission, and companies cannot refuse service just because someone says no. Apps and website must be explained in simple language what data they are collecting and why. Smart devices sold in Vits must pass a privacy safety check before they can be sold in stores. Companies that illegally collect or sell personal data should face double the normal fine especially if biometric data is involved.

Digital Identity

Today in VIT's citizens use different username and password for different government ,websites banks and apps. This is not only inconvenient but also unsafe, because weak password leads to identity theft. VIT'S needs one secure and unified digital identity system that citizens can use everywhere without having to share sensitive documents repeatedly.

This system should be designed so that Citizens Store their own identity on their personal device, not on one of the central government server. This safer because if one database is hacked the entire country's identity data is not exposed. People who do not have smartphone should get a physical card that works the same way. The system should have different security levels - Basic for general websites and stronger verification for banking or legal matter. The NCSA Should check the security of this system every year,and if There is a breach,all affected citizens must be informed within 24 hours

Cyber Warfare

Cyber crime doesn't always come from individual hackers. Sometimes, Foreign government secretly attack another country's system to steal information , disrupt elections or damage infrastructure. This is called cyberwarfare And it is one of the most serious future threads for VIT's. The current law mainly deals with individual criminals and is not designed to handle Govt - backed attacks.

VIT should define cyber warfare clearly in the law. Any attack on VIT's power grid,election system, military network or water supply that is linked to a foreign government should be treated as an act of war in the digital space. When such an attack happens, The government should be allowed to declare a national cyber emergency and take a quick action. Vit should also build a separate cyber defence command under the Defence Ministry that focuses specifically on responding to and deterring state-sponsored attacks. Before responding to any such attack, Vits must gather solid evidence and follow international rules to avoid any misunderstanding or unnecessary conflicts.



Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

Criteria	Max Marks	Marks Obtained
Research Quality — depth and accuracy of information, use of real-world examples	20	
Law Structure — completeness, logical organization, and legal clarity of the law	20	
Innovation — originality of solutions, creative approach to cybersecurity challenges	15	
Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures	15	
Presentation — quality of the PowerPoint presentation and oral delivery	10	
Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables	10	
Case Studies — realism, detail, and correct application of the law	10	
TOTAL	100	

Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

Don	Deliverable	Requirements
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[]	Project Report	Minimum 25 pages, well formatted, all 15 parts completed, diagrams included
[]	PowerPoint Presentation	15 to 20 slides, covering all major sections of the law
[]	Flowchart: Cybercrime Reporting	Shows complete incident reporting process from detection to resolution
[]	Flowchart: Investigation Process	Shows NCSA investigation steps from complaint to judgment

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Don e	Deliverable	Requirements
[]	Flowchart: Data Breach Response	Shows data breach response procedure for organizations
[]	Infographic: Citizens' Rights	Visual summary of all citizens' digital rights
[]	Infographic: Cybercrime Categories	Visual taxonomy of all defined cybercrimes with penalties
[]	Infographic: Security Framework	Visual overview of the data protection and corporate security framework

Good Luck!

This project challenges you to think like a lawmaker, a technologist, and a citizen. Research real world cybersecurity laws (India's IT Act, GDPR, Budapest Convention) for inspiration. Be creative, be thorough, and make VIT's digital future secure!

